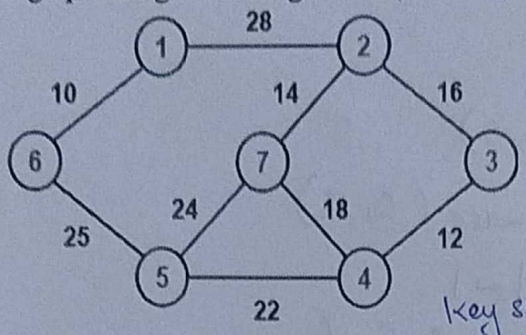
ASSIGNMENT - I

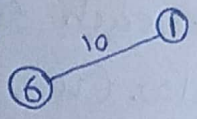
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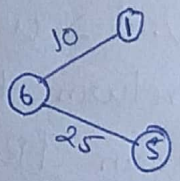
4. Explain the different steps used to find the minimum cost spanning tree for the below graph using Prim's algorithm.(6 Marks)



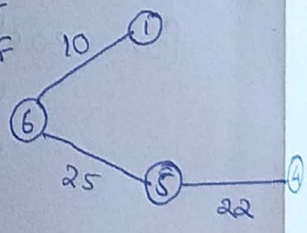
Initially,
Keyset = {0, INF, INF, INF, INF, INF, INF}
mstSet = {1}
from 1 ⟶ (1,2) = 28 < INF
(1,6) = 10 < INF
Keyset = {0, 28, INF, INF, INF, 10, INF}
mstSet = {1, 6}



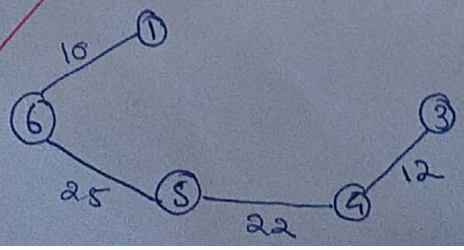
from 1 ⇓ (1,2) = 28 = 28
from 6 ⇓ (6,5) = 25 < INF
Update keyset
minimum (6,5) ⟹ add mstSet
∴ Keyset = {0, 28, INF, INF, 25, 10, INF}
mstSet = {1, 6, 5}



from 1 ⇓ (1,2) = 28 = 28
from 6 ⇓ (6,5) = 25 = 25
from 5 (5,7) = 24 < INF
(5,4) = 22 < INF
Update keyset
∴ Keyset = {0, 28, INF, 22, 25, 10, INF}
mstSet = {1, 6, 5, 4}



from 1 ⇓ (1,2) = 28 = 28
from 6 ⇓ (6,5) = 25 = 25
from 5 (5,4) = 22
from 4 (4,7) = 18 < INF
(4,3) = 12 < INF
Update keyset
∴ Keyset = {0, 28, 12, 22, 25, 10, 18}
mstset = {1, 6, 5, 4, 3}

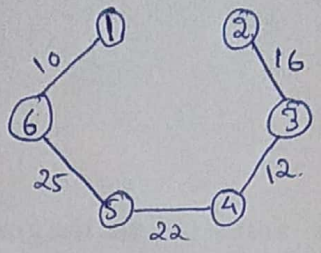


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from 4, from 3,

(4,3) = 12 (3,2) = 16 < 28

update keyset



pick (3,2) minimum,

∴ keyset = {0, 16, 12, 22, 25, 10, 18}

mstSet = {1, 6, 5, 4, 3, 2}

from 2

(2, 7) = 14 < 18

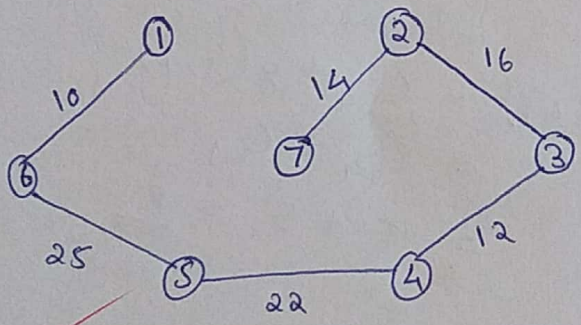
update keyset

∴ keyset = {0, 16, 12, 22, 25, 10, 14}

mstSet = {1, 6, 5, 4, 3, 2, 7}

all vertices are available on mstset so stop the process.

Minimum spanning tree using prims algorithm is,



Total weight ⟹ 10 + 25 + 22 + 12 + 16 + 14

= 99

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